COGNITIVE·TERRAIN

BLUE vs RED · THE WAR FOR HOW A NATION THINKS



NO SHOTS FIRED. SIXTEEN WEEKS TO WIN A MIND.

The complete rules, every screen explained, and the research behind it.

8 MOVES DECODED

Plant Stories to Lessons Learned: what each card does and where it hits hardest.

SWAP CODES, NOT SERVERS

Play a friend over email or chat. No accounts needed.

EDUCATIONAL WARGAME · TURN-BASED
WINDOWS · MAC · LINUX · ANY BROWSER

CT · 0017 · 2026

INSIDE THIS MANUAL

CONTENTS

03 **1.0 INTRODUCTION**
What the game is, how you win, what it is not

04 **2.0 THE SCREEN AND THE BOARD**
The two minds, the five areas, every control numbered

08 **3.0 SETTING UP**
Campaign, difficulty, terms and the tutorial

09 **4.0 SEQUENCE OF PLAY**
One week, in order

10 **5.0 CONTROL**
Leaning, footholds, holding and losing areas

11 **6.0 BUDGET AND MOVES**
Income, costs, the extra move, making a move

13 **7.0 ATTACK MOVES**
Reach, the four attacks, push, failure and the spread grade

18 **8.0 DEFENCE MOVES**
The four defences and the statuses they set

20 **9.0 UPKEEP**
Recovery, spread and the week in review

22 **10.0 CRISES**
The four moments that score

23 **11.0 NEW TECHNOLOGIES**
Use it or prepare for it

24 **12.0 WINNING**
Points, ties, the knockout and the debrief

25 **13.0 THE OPPONENTS**
Blue, Olvana and Donovia

28 **14.0 TWO PLAYERS**
Turn codes and White Cell sessions

30 **15.0 SETTINGS**
Accessibility and Immersive Briefings

32 **APPENDICES**
A Tables · B Glossary · C Player aid · D Sources

HOW TO READ THIS MANUAL

`5.0`	A section. Sections follow the order of a week of play.
`5.3`	A rule. Each rule is stated once, in its own section.
`[5.3]`	A pointer to that rule from somewhere else.
`EXAMPLE`	A rule worked through with real numbers.
`PLAYER'S NOTE`	Advice in one line. Notes are not rules.
`CAPITALS`	Words you will see on screen, such as END TURN.

This manual uses the game's plain terms. Appendix B gives the military term for each.

FIRST GAME?

Play **Typhoon Ines** on **Recruit** and press **START WITH TUTORIAL**. It takes about ten minutes. Then read 4.0 and keep the player aid (Appendix C) beside you.

Running it for a group, or building your own campaign? The separate **Practitioner's Manual** covers the campaigns, sessions, the White Cell, the debrief, the editor and the designer's notes behind each rule.

SECTION 1 · READ THIS FIRST

1.0 INTRODUCTION

Cognitive Terrain is a game about deciding under pressure. You win by keeping your own side's judgment clear while clouding the other side's.

1.1 WHAT THE GAME IS ABOUT

Countries compete in many ways short of war. One of them is reaching into the other side's newsfeeds, group chats and briefing rooms to change what people notice, what they believe and how fast their leaders decide.

The game turns that contest into a board you can play over a lunch break. The board is two minds, one for each side. Each mind is a network of nodes, and each node is a kind of person, group or institution: a newsroom, a cabinet, a crisis team, the public's trust in its own government.

You command **Blue**, an open democracy. Your opponent is **Red**: Olvana, a closed, centralized state, or Donovia, an oligarchic one. The campaign you pick decides which. A campaign lasts sixteen weeks.

1.2 HOW YOU WIN

Four crises score points [10.0]. The side with more points after the last crisis wins [12.1]. A side can also win early by knockout [12.3].

Nodes, posts and audiences do not score by themselves. A crisis scores how well each side decides at that moment, which depends on the health of its mind.

> PLAYER'S NOTE Reach is not effect: aim at the area the next crisis tests.

1.3 WHAT THIS GAME IS, AND IS NOT

> Cognitive Terrain is an independent hobby project, made for learning and discussion. It is **not doctrine**, **not an official or endorsed product**, and its numbers have **not been independently validated**. It simplifies published research into game rules; where a rule is the designer's own assumption, the Practitioner's Manual says so. Every campaign, town, outlet, person and event in it is fiction.

1.4 THE TWO MANUALS

This **Field Manual** is for players. It is also the guide inside the game, so you can look up a rule without leaving a match.

The **Practitioner's Manual** is for facilitators and designers: the eight campaigns in full, running a session, the White Cell, the debrief, the model in numbers, the campaign editor and the designer's notes, each keyed to a rule number in this manual.

1.5 ROUNDING

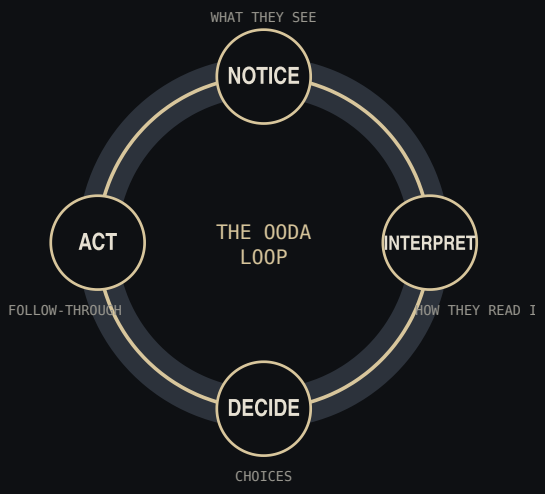
The game keeps exact values and shows rounded ones. A health of 0.857 shows as 0.86 but still counts as under 0.86. When a number here and a number on screen seem to disagree by a hair, rounding is the reason.



FIG. 0 Part of Olvana's mind in a campaign. Each dot is a node; faint words name the areas. Blue has pushed Olvana's Social Media 41 its way and now holds it.

SECTION 2 · WHAT YOU ARE LOOKING AT

2.0 THE SCREEN AND THE BOARD



2.1 THE TWO MINDS

The board shows two minds, Blue's and Red's. Each is a network of 30 nodes joined by links. You attack the other side's mind and defend your own.

Every mind runs the same decision loop. Picture a town hearing that a storm is coming: people **notice** the warning, **interpret** it, leaders **decide**, and people **act**. Under all of it sits **trust**. Planners call the four steps the **OODA loop**: Observe, Orient, Decide, Act.

2.2 NODES AND ENTRY POINTS

Each node belongs to one of five areas [2.3]. Links show who influences whom. Each mind has one **Top Leadership** node, drawn large, and three **entry points** ◇ : **News Media** (in NOTICE), **Government Contacts** (in INTERPRET) and **Social Media** (in TRUST). Entry points are where outside messages get in [7.1].

2.3 THE FIVE AREAS

AREA	WHAT IT IS IN REAL LIFE	WHY IT MATTERS IN THE GAME
NOTICE	Where people get information: news, social feeds, reports from the ground, intelligence briefings.	Holds News Media. Losing it clouds your view [5.5].
INTERPRET	How people make sense of it: who is to blame, what it means, what it looks like from the past.	Holds Government Contacts. Losing it makes every move dearer.
DECIDE	Leaders and teams who weigh options and choose: cabinets, planners, appetite for risk, red lines.	Holds Top Leadership. Losing it costs a move a turn.
ACT	Carrying decisions out: troops, supplies, public cooperation, keeping promises.	Losing it makes moves misfire.
TRUST	The slow foundation: trust in institutions, unity, shared identity, knowing where to check facts.	Holds Social Media. Scales the whole of Decision Health [10.6].

PLAYER'S NOTE TRUST falls slowly and costs the most when it goes; guard it from week 1.

SECTION 2 · ANATOMY OF THE WAR ROOM

2.4 THE SCREEN, NUMBERED

The big mind in the middle is the one you are looking at; the small one, top right, is the other. Click either tab or the small mind to swap.

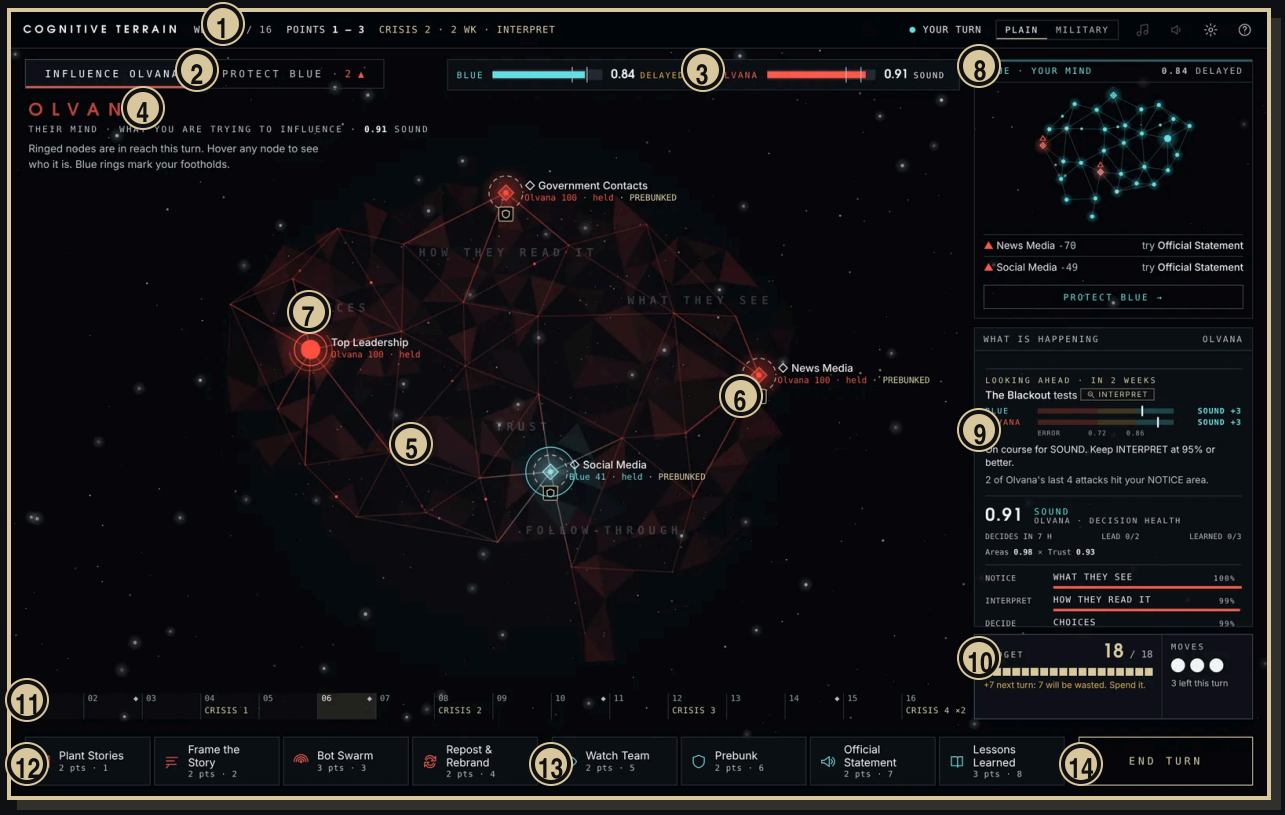


FIG. 2 Week 6 of a Typhoon Ines campaign, looking at Olvana's brain. The top-right panel lists Blue's own nodes under most pressure, starting with Social Media; below it, Looking ahead previews the next crisis.

1 TOP BAR
Week, crisis points so far (Blue – Olvana), and the next crisis with the area it will test.

2 VIEW TABS
INFLUENCE OLVANA shows their brain. PROTECT BLUE shows yours; the ▲ count is how many of your nodes are under pressure.

3 HEALTH BARS
Each side's Decision Health, 0 to 1, with marks at the crisis cut-offs. Label reads SOUND, DELAYED or ERROR.

4 BRAIN TITLE
Whose brain this is, its health, and a one-line hint about what you can do here right now.

5 THE BRAIN
Nodes are people, groups and institutions; lines are who influences whom. Faint area names mark the five regions.

6 ENTRY POINT ◇
News Media, Social Media and Government Contacts. Every attack starts here or next to a node you already lean.

7 TOP LEADERSHIP
The big core node. Worth 2 budget a turn to whoever holds the other side's.

8 THE OTHER BRAIN
A live mini-map. On your own brain it lists the nodes under pressure and suggests a defence.

9 WHAT IS HAPPENING
Looking ahead to the next crisis, health, the five areas and the event log.

10 BUDGET & MOVES
Points to spend and moves left this turn. Hover a card to see what it would cost.

11 TIMELINE
The sixteen weeks. ◆ marks a new technology, CRISIS the four decision points. Above it, this week's orders stack up as icons; click one for its story.

12 ATTACK CARDS
Four moves that push the other brain. Keys 1 to 4.

13 DEFENCE CARDS
Four moves that protect your own brain. Keys 5 to 8.

14 END TURN
Pulses when you are out of moves or budget. In two-player games it hands you a turn code.

SECTION 2 · CLOSE-UPS

2.5 THE INSTRUMENTS, UP CLOSE

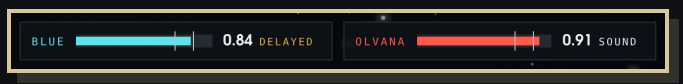


FIG. 3 Health bars. Blue on the left, Olvana on the right. The tick marks are the crisis cut-offs: above the right tick a crisis goes SOUND, between the ticks DELAYED, below them ERROR.



FIG. 4 View tabs and brain title. The "2 ▲" on PROTECT BLUE says two of your nodes are being pushed. The hint line changes with the card you pick.



FIG. 5 Budget and moves. 18 of 18 points, 3 moves left, and +7 arriving next turn. The budget is full, so all 7 would be wasted: spend it.

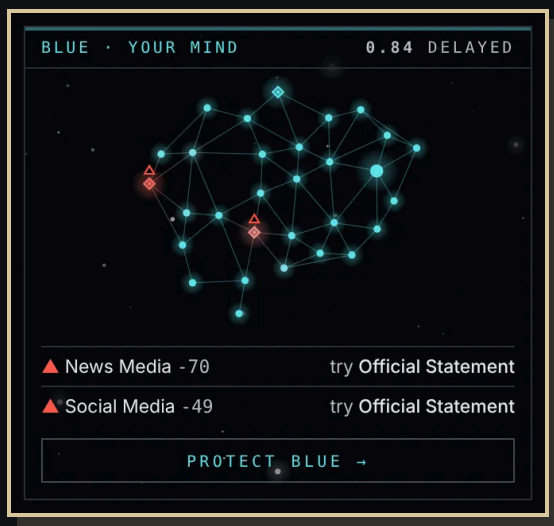


FIG. 6 The mini panel while you view Olvana. It watches your home brain for you: each ▲ row names a node under pressure and the card that would help. When one more loss would cost you an area, a red-ruled warning appears here too.

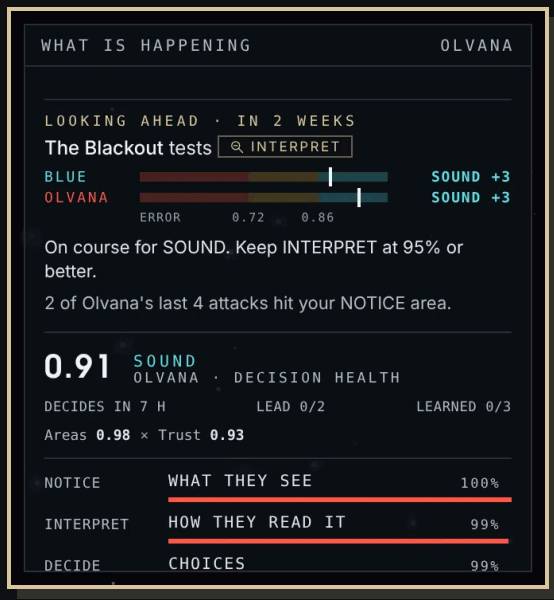


FIG. 7 What is happening. At the top, Looking ahead: the next crisis, how each side would handle it today and what would change that. Below: health, how many hours a decision takes, the race to the upper hand (LEAD 0/2) and how much of each area the brain still controls.

SECTION 2 · FIELD GUIDE TO THE MIND

2.6 MAP SYMBOLS

Colour tells you who a node leans toward; rings and small icons tell you what has been done to it.

SYMBOL KEY

Cyan node. Leans Blue. Brighter means stronger.

Red node. Leans Olvana.

Amber dotted ring. Contested: nobody holds it.

Status icons under a node. **Eye**: watched (attacks land weaker, your statements hit harder). **Shield**: prebunked (incoming pushes count half). **Lines**: framed (it will not recover for a while). Status words on its label use the same colours.

Dashed outline. One Watch Team or Prebunk covering a node and its links, tagged with weeks left. Pick the card and hover a node to preview what it would cover.

Large node. Top Leadership, the core of the brain.

▲ on your brain. Under pressure. Defend it soon.

HIDDEN BY DESIGN

Only entry points and Top Leadership are named all the time. Hover any node to see who it is. Settings can show every name [15.1].



FIG. 8 Olvana's mind at week 6. Three entry points (◇) and Top Leadership are labelled. Olvana has prebunked its entry points and put a watch team on Social Media. Blue has no foothold yet. The faint words are the area names.

2.7 WHAT THE COLOURS MEASURE

A node's colour shows its **lean**, the control score in 5.1. How far it leans decides whether a side has a foothold there or holds it [5.2] [5.3].

> PLAYER'S NOTE The ▲ count on PROTECT BLUE is the fastest check on your own mind.

SECTION 3 · BEFORE WEEK 1

3.0 SETTING UP

3.1 CAMPAIGN

Pick one of eight fictional campaigns. The campaign sets Red's country, the four crises and the areas they test, the new technologies and any special starting conditions. The Practitioner's Manual (3.0) describes each in full.

CAMPAIGN	RED	IN ONE LINE
TYPHOON INES	Olvana	Disaster relief on an island republic. No twist: the best first game.
THE VALMORA VOTE	Olvana	A snap election. Both TRUST areas start weakened.
VARRIN REEF	Olvana	A collision at sea. Both Government Contacts start weakened.
PORT SABRE	Olvana	An outbreak. Strongly held nodes push harder on their neighbours.
WINTER GRID	Donovia	A cold snap. Blue's ACT starts strained; 4 dormant narratives.
LAKE REFERENDUM	Donovia	A dam vote. Donovia starts with a foothold in Blue's INTERPRET.
SILENT SURVEY SHIP	Donovia	A ship goes quiet. Both DECIDE areas start strained.
THE VELL VALLEY	Donovia	An annexation. 5 dormant narratives; Blue's NOTICE and TRUST strained.

3.2 DIFFICULTY

Recruit: the computer is slower and sloppier, and Red gets 2 fewer points a turn. **Standard**: a steady, sensible opponent. **Veteran**: it plans two moves ahead and plays to win, with no extra budget and no hidden information.

3.3 INSTALLING

SYSTEM	WHAT TO DO
ANY PC	Open the .html file in a browser. Nothing to install; no account; works offline.
WINDOWS	Run the Setup .exe. If Windows says it "protected your PC", choose *More info* → *Run anyway*: the playtest build is not code-signed.
MAC	Unzip, drag to Applications, then right-click → *Open* the first time.
LINUX	Unpack the .tar.gz and run `./cognitive-terrain`.



FIG. 1 The start screen. Campaign, language, difficulty and sound are all set here before week 1.

3.4 PLAIN OR MILITARY TERMS

Plain uses everyday words ("Decision Health", "Plant Stories"); **Military** uses doctrine terms ("DQ", "Seed Channels"). Switch any time; the rules do not change [App. B].

3.5 STARTING

Press **START WITH TUTORIAL** or skip straight in. **CAMPAIGNS** opens the editor to make or import a campaign; **TWO PLAYERS** starts a game by turn codes [14.0].

SECTION 4 · ONE WEEK, IN ORDER

4.0 SEQUENCE OF PLAY

A campaign is sixteen weeks. Every week runs through the phases below, in this order.



FIG. 13 The timeline. ◆ marks a new technology; CRISIS 1–4 fall in weeks 4, 8, 12 and 16, and crisis 4 counts double (×2). Above it, this week's moves stack up as icons; click one for its story.

4.1 NEW TECHNOLOGY PHASE

Weeks 2, 6, 10 and 14 only. Both sides choose in secret to use the new technology or prepare for it [11.0].

4.2 BLUE'S TURN

Blue collects income [6.1], makes its moves [6.2] and presses **END TURN**.

4.3 RED'S TURN

Red does the same. Each Red move lights up on your map, numbered, with a small card saying what it did and why.

4.4 UPKEEP

Nodes recover, strong nodes push on their neighbours, and statuses count down [9.0].

4.5 CRISIS PHASE

Weeks 4, 8, 12 and 16 in the built-in campaigns. Both sides are scored [10.0].

4.6 WEEK IN REVIEW

Both sides' moves, what changed and what comes next [9.6]. Then the next week begins.

4.7 END OF THE GAME

The game ends after the crisis in week 16, or at once when a side wins by knockout [12.3]. The debrief follows [12.4].

4.8 TWO PLAYERS AND THE WHITE CELL

In a two-player game each turn ends with a code for the other player [14.1]. In a structured session the White Cell acts between Red's turn and Blue's next turn [14.6].

WATCH THE CLOCK

The top bar always names the next crisis and the area it tests, for example `CRISIS 2 · 2 WK · INTERPRET`.

EXAMPLE

Week 4 of Typhoon Ines. No new technology this week. Blue moves, then Olvana moves. Upkeep runs. Then the first crisis, The Relief Flotilla, tests NOTICE and both sides are scored. The week in review shows what happened, and week 5 begins.

PLAYER'S NOTE Upkeep runs before the crisis, so a crisis week's recovery counts toward the score.

5.0 CONTROL

5.1 LEAN

Every node has a control score from −100 (fully Red) to +100 (fully Blue). The score is the node's **lean**. Nodes start leaning strongly toward their owner; some campaigns start a few nodes weaker [3.1].

5.2 FOOTHOLD

A side has a **foothold** on a node that leans 10 or more its way. From a foothold, that side's attacks can reach every node linked to it [7.1].

5.3 HELD

A side **holds** a node that leans 30 or more its way. A held node in the enemy mind earns income [6.1] and does not recover on its own [9.1]. A node leaning less than 30 either way is contested.

5.4 LOSING AN AREA

An area is **lost** for its owner while the other side holds at least half of its nodes (rounded up: 3 of 5, 3 of 6). DECIDE is also lost while the other side holds its Top Leadership. An area that is no longer held that way is won back at once.

EXAMPLE

Olvana's INTERPRET area has 6 nodes. Blue holds 2 of them: not lost. Blue pushes a third to 30: half of 6 is 3, so INTERPRET is lost for Olvana, and every Olvana move now costs 1 more. If Olvana pulls one back under 30, the area is won back.

5.5 WHAT EACH LOST AREA DOES

These penalties apply to the side that lost the area, for as long as it stays lost.

AREA LOST	PENALTY
NOTICE	Too much noise: you can no longer see the other side's mind clearly.
INTERPRET	Every move costs 1 more point.
DECIDE	1 fewer move per turn.
ACT	1 in 4 of your moves misfires and does nothing; you still pay for it.
TRUST	Your Official Statements lose their bonus and are halved (Blue's fall from 22 to 7) and are ignored more often; your recovery halves; enemy posts into your mind reach ten times the audience.

Every lost area also takes 0.06 off your Decision Health [10.6].

5.6 SEEING IT ON SCREEN

The side panel shows how much of each area a mind still controls. When an area is one node from falling, a red-ruled warning names the penalty [8.6].

> PLAYER'S NOTE A third node is worth more than the first two: it is the one that takes the area.



The lean scale [5.1]–[5.3]. A held node also gives its holder a foothold.

6.0 BUDGET AND MOVES



FIG. 9 Three moves and a full budget of 18 points.



FIG. 10 Out of moves with at least 4 points left: the +1 MOVE button appears [6.4].

EXAMPLE

Blue holds three Olvana nodes, one of them Government Contacts. Income is 6 + 3 + 2 = **11**. Blue has 9 points left over, so it would reach 20; the budget stops at 18 and 2 points are lost.

PLAYER'S NOTE Near 18 points? Spend down or surge before you end the turn.

6.1 INCOME

Each side starts with 8 points. At the start of each of its turns, a side adds:

SOURCE	POINTS
Base income	+6
Each node you hold in the enemy mind	+1
Their Top Leadership, if held (instead of +1)	+2
Their Government Contacts, if held (on top of its +1)	+2

Income is never less than 2. Some opponents and difficulty levels change Red's base income [3.2] [13.4].

6.2 MOVES PER TURN

Each side makes up to 3 moves a turn, or 2 while its DECIDE area is lost [5.5]. Each move plays one card. You may end your turn with moves or points unspent.

6.3 COSTS

Most cards cost 2 points; Bot Swarm and Lessons Learned cost 3. Add 1 to every move while your INTERPRET area is lost, and 1 to your first move after a Donovian Show of Force [13.4]. The cost on each card already includes these.

6.4 THE EXTRA MOVE

Once a turn, after your last move, you may buy one extra move: Blue pays 2 points, Red pays 5. You must have at least 2 points left after paying, so Blue needs 4 and Red 7.

6.5 THE 18-POINT CAP

A side can hold at most 18 points. Income that would take it past 18 is lost.

SECTION 6 · ONE MOVE FROM START TO FINISH

6.6 MAKING A MOVE

Pick a card, find a node it can reach, preview, then click to commit. Esc cancels at any point before the click.



STEP 1 · PICK A CARD Click a card in the bottom bar or press its key (1 to 8; 9 is Donovia's Show of Force). Attacks are on the left, defences on the right, each with its cost. Picking an attack shows the enemy mind; a defence shows yours.



STEP 2 · FIND A RINGED NODE With Plant Stories selected, the nodes this card can reach are ringed [7.1].



STEP 3 · HOVER TO PREVIEW The hover card shows who the node is, who holds it, the typical push, the chance to fail and the cost. The side panel adds the bonuses and the real-life reasons. Click to commit.



STEP 4 · SEE WHAT HAPPENED The headline, a few lines of the story, the grade [7.6] and what changed. FULL STORY opens the complete report.

6.7 COMMITTING

A move is final once you click the node. Its cost is paid even if it fails [7.5] or misfires [5.5].

6.8 CARDS WITHOUT A TARGET

Lessons Learned needs no node: click the card and it fires [8.1].

> PLAYER'S NOTE Hover before you click: the preview already counts every bonus and status.

SECTION 7 · KEYS 1–4

7.0 ATTACK MOVES

7.1 REACH

An attack targets one node in the enemy mind. The node must be an **entry point**, a node where you have a **foothold** [5.2], or a node **linked** to a foothold. Nodes in reach are ringed when the card is picked. Two exceptions: Show of Force [13.4] and Humanoid Robots [11.3].

7.2 THE FOUR ATTACKS

PLANT STORIES

KEY 1

GET THE MESSAGE IN · MILITARY: SEED CHANNELS

COST	PUSH	TARGETS	SPECIAL
2	20	1	×1.5 ◇

NOTICE ×1.5 | INTERPRET | DECIDE ×.75 | ACT ×.75 | TRUST

Place content in the news and feeds people already use. The strongest base push in the game, and 50% stronger on an entry point. Your opening move.

> IN REAL LIFE: look-alike local news sites, sponsored posts, paid influencers.

FRAME THE STORY

KEY 2

SHAPE THE STORY · MILITARY: FRAME & PRIME

COST	PUSH	TARGETS	SPECIAL
2	16	1	FRAMED

NOTICE ×.75 | **INTERPRET ×1.5** | **DECIDE ×1.25** | ACT | **TRUST ×1.25**

Tell it in a way that triggers emotion or identity. For 2 turns the target cannot recover and every enemy attack on it lands 40% harder.

> IN REAL LIFE: a clip edited and captioned to make people angry or afraid.

BOT SWARM

KEY 3

MAKE IT LOUD · MILITARY: COORDINATED AMPLIFICATION

COST	PUSH	TARGETS	SPLASH
3	12	+LINKS	7 EACH

NOTICE | INTERPRET | DECIDE | **ACT ×1.5** | **TRUST ×1.25**

Many coordinated accounts repeat it at once. Hits the target *and every node linked to it*. Hold their Social Media entry point and the splash hits at full strength.

> IN REAL LIFE: thousands of accounts posting the same thing within an hour.

REPOST & REBRAND

KEY 4

ADAPT · MILITARY: PIVOT & RESEED

COST	PUSH	TARGETS	STRIPS
2	10	1	+LINKS

NOTICE | INTERPRET | DECIDE | ACT | TRUST

Come back with new accounts and a new angle after removals. Strips **Watched** and **Prebunked** from the target and its links. Use it when the other side digs in.

> IN REAL LIFE: a banned network reappearing under new names the next day.

PLAYER'S NOTE Frame a node, then plant on it next turn: it cannot recover and takes 40% more.

7.3 AREA FIT AND PUSH

7.3 AREA FIT

Each attack works better in some areas of the mind than others. A card's push is multiplied by its fit for the target's area:

ATTACK	NOT	INT	DEC	ACT	TRU
PLANT STORIES	×1.5	×1	×0.75	×0.75	×1
FRAME THE STORY	×0.75	×1.5	×1.25	×1	×1.25
BOT SWARM	×1	×1	×1	×1.5	×1.25
REPOST & REBRAND	×1	×1	×1	×1	×1

Highlighted values are each card's best areas. Defences work the same in every area.

> PLAYER'S NOTE Match the card to the area: Frame the Story in INTERPRET, Bot Swarm in ACT.

7.4 THE PUSH

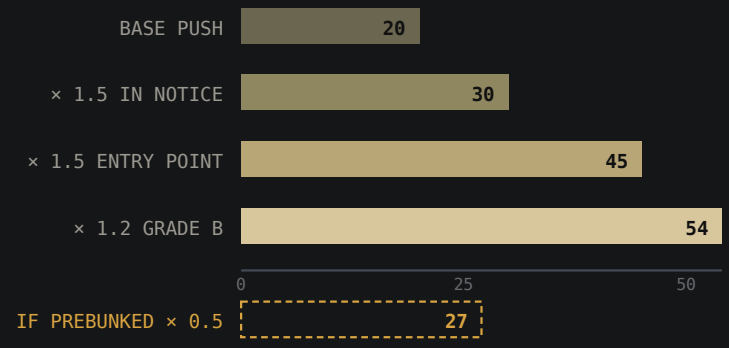
When an attack works, it moves the target node toward you. How far depends on four things, each of which makes the push bigger or smaller:

- **The card.** Each attack has a base push: Plant Stories 20, Frame the Story 16, Bot Swarm 12, Repost & Rebrand 10 [7.2].
- **The area.** A card hits harder in the areas it fits best [7.3].
- **The audience.** A post that reaches and engages more people pushes harder; a flop pushes less. The grade sets this, from a quarter of the push to 60% more [7.6].
- **The node's state.** A prebunked node takes only half; a framed node takes 40% more [8.2].

Plant Stories also lands 50% harder on an entry point. Bot Swarm, as well as hitting its target, pushes every linked node by 7 (12 while you hold their Social Media, or with a technology that gives full splash [11.3]), adjusted the same way.

EXAMPLE

Blue plays **Plant Stories** on Olvana's **News Media**, an entry point in NOTICE. The post earns grade **B**. Each step makes the push bigger: 20 to start, 30 for the best area, 45 for the entry point, **54** for the good audience. Had Olvana prebunked News Media, the same post would push only **27**.



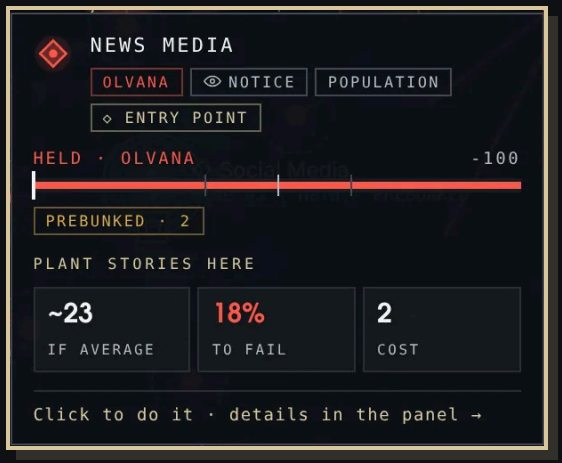


FIG. 7.4 · THE SAME MOVE IN THE GAME
You never need to work this out yourself. Pick a card and hover a node: the card shows the push you would get with an average audience, every bonus already counted.

Here News Media **is** prebunked, so Plant Stories shows **~23**: 20 × 1.5 × 1.5 × 0.5 = 22.5. The **18%** to fail is 8 + 8 for the prebunk + 2 for Olvana's censors [7.5].

7.5 FAILURE AND BACKFIRE

7.5 FAILURE AND BACKFIRE

Before it pushes, every attack has a chance to fail: platforms take it down, watchers expose it, or people shrug it off. The chance starts at 8% and rises when the target is defended:

CONDITION	CHANCE
Any attack	8%
Target watched	+15%
Target prebunked	+8%
Blue posting into Olvana / Donovia	+2% / +1%
Show of Force	+4%

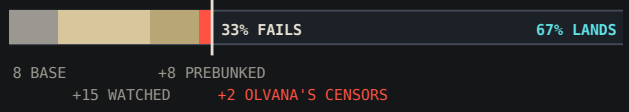
The hover card shows this chance before you commit (FIG. 7.4).

A failed attack does not just miss: it pushes its target 6 toward the defender (9 when a Donovian attack fails) [13.4]. The cost is still paid.

Official Statements can fail too: they are ignored 6% of the time, 26% while your TRUST area is lost. An ignored statement does nothing.

EXAMPLE

Blue plants a story on an Olvana node that is both watched and prebunked. The chance it fails is 8 + 15 + 8 + 2 = **33%**, about one in three. If it fails, the node moves 6 further toward Olvana.



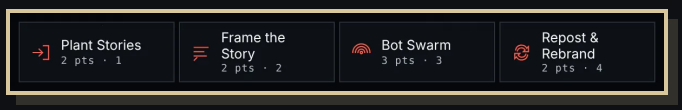


FIG. A The attack half of the card bar. A card you cannot afford is dimmed.

PLAYER'S NOTE Repost & Rebrand clears a watched, prebunked cluster for the attacks that follow it.

7.6 AUDIENCE AND GRADE

Every attack except Show of Force is a small campaign of posts. The game rolls the audience it reached, then grades the campaign the way the research does.

SPREAD SCORE PER POST

$$(0.1 \cdot \text{views} + 0.3 \cdot \text{likes} + 0.7 \cdot \text{comments} + 1 \cdot \text{shares}) \div \text{posts}$$

RUSHING & XU (2025) · ENGAGEMENT EFFECTIVENESS

GRADE	SCORE	MEANING	EFFECT
A+	10,000+	Viral	×1.6
A	1,000+	Excellent	×1.4
B	100+	Above average	×1.2
C	10+	Average	×1.0
D	4 to under 10	Below average	×0.7
E	3 to under 4	Poor	×0.45
F	under 3	Failure	×0.25

Each attack has its own crowd. **Plant Stories** draws lots of views and few shares. **Frame the Story** sparks the most comments. **Bot Swarm** racks up shares but little real conversation.



FIG. 12 The full story (◉ in the log, or FULL STORY on a result card): audience counts with their weights, the grade and its formula, what changed and why it matters.

What decides how big the audience is, and so the grade, is in 7.7.

> PLAYER'S NOTE A viral post on the wrong area still loses the crisis: aim first.

7.7 WHAT MOVES THE AUDIENCE

Every attack's audience is rolled, so the same move can flop or go viral. But the odds are not even: where you post, what you hold and what the defender has done all shift the roll up or down.

7.7 WHAT MOVES THE AUDIENCE

Each attack has a typical audience. The things below multiply it. Because the grades go up in steps of ten (10, 100, 1,000, 10,000), anything that multiplies the audience by ten usually moves the post **one grade** up or down [7.6].

WHAT HAPPENS	AUDIENCE	WHY
A technology you chose to use	×10	New tools reach and target more people, unless the other side prepared [11.2].
Their TRUST area is lost	×10	People no longer trust their own sources, so rumours travel.
You hold their News Media	about ×3	Your posts borrow a trusted outlet's reach.
Posting on an entry point	×2.5	You are where the audience already is.
Red posting into Blue	×1.25	Blue's open internet carries it further.
Blue posting into Olvana / Donovia	÷1.4 / ÷1.25	Censors filter it; Donovia's filter is leakier.
A Watch Team on the target	÷10	Spotted early, it is taken down before it spreads [8.2].

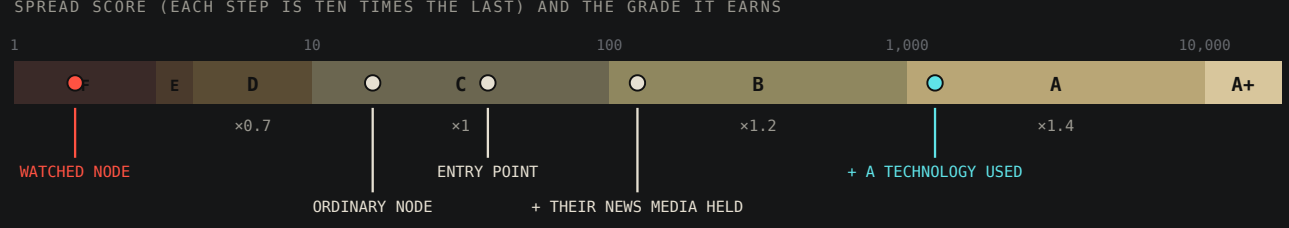
READING IT IN PLAIN WORDS

- **The big three** are a technology, a lost TRUST area and a Watch Team. Each is worth about a whole grade.
- **The small ones** (entry point, News Media, open or censored internet) add up: two or three together are worth about a grade.
- **Luck still matters.** The roll often lands a grade above or below the typical one. The hover card shows the push for an average audience [7.4].

> PLAYER'S NOTE Before a big attack, check for a Watch Team: it costs you a whole grade.

EXAMPLE

The same Plant Stories, five places. Blue's typical post into Olvana scores about **16**: grade **C**, push ×1. On an entry point it is about 39, still C. If Blue also holds Olvana's News Media, about 124: **B**, ×1.2. Add a technology Blue chose to use: about 1,240, **A**, ×1.4. But the same post on an ordinary node under a Watch Team scores about 1.6: **F**, ×0.25, a quarter of the push.



SECTION 8 · KEYS 5–8

8.0 DEFENCE MOVES

Defences go on your own mind. Attacking is cheap and defending is not, so defences work best when they are in place before the attack arrives.

8.1 THE FOUR DEFENCES

WATCH TEAM

KEY 5

SPOT IT · MILITARY: ASSESSMENT CELL

COST	LASTS	TARGETS	BOOST
2	3 WK	+LINKS	×1.5

Same in every area of your mind

Watches a node and its links. Incoming posts reach a tenth of the audience (usually a grade or more weaker) and fail far more often (23% instead of 8%), framing is cleared, and your Official Statements there hit 50% harder.

> IN REAL LIFE: analysts tracking sudden spikes of identical posts.

PREBUNK

KEY 6

BUILD IMMUNITY · MILITARY: PREBUNK & INOCULATE

COST	LASTS	TARGETS	INCOMING
2	3 WK	+LINKS	×0.5

Same in every area of your mind

Teach people the trick before they see it. A whole cluster takes half influence for three weeks. Best laid down *before* the pressure arrives.

> IN REAL LIFE: short videos showing how fake clips are made, shown before an election.

OFFICIAL STATEMENT

KEY 7

SET THE RECORD STRAIGHT · MILITARY: AUTHORITATIVE RESPONSE

COST	RESTORE	TARGETS	IF WATCHED
2	14+	1	×1.5

Same in every area of your mind

Correct the record through trusted voices. Pulls one node back toward you. Blue's free press makes it hit extra hard while Blue's TRUST holds, and it falls to 7 once TRUST is lost.

> IN REAL LIFE: a press conference with evidence from a trusted spokesperson.

LESSONS LEARNED

KEY 8

LEARN & PRACTICE · MILITARY: AFTER-ACTION REVIEW

COST	ALL NODES	TARGET	MAX
3	+4	NONE	3×

Same in every area of your mind

Review what went wrong and drill. Every node you own gains 4, framing is cleared, and your weekly recovery gets permanently faster (up to three times). Pays more the earlier you buy it.

> IN REAL LIFE: an after-action review followed by a practice exercise.

PLAYER'S NOTE Watch Team then Official Statement on the same node: the statement lands 50% harder.

8.2 WATCHED, PREBUNKED, FRAMED

STATUS	SET BY	LASTS	EFFECT ON THE NODE	ENDED EARLY BY
WATCHED	Watch Team, on the node and its links	3 weeks	Enemy posts reach a tenth of the audience and fail 15% more often; your Official Statements there ×1.5; a Donovian dormant narrative there is removed.	Enemy Repost & Rebrand
PREBUNKED	Prebunk, on the node and its links; preparing for a new technology, on your entry points	3 weeks (2 when preparing)	Enemy pushes ×0.5 (×0.45 against Donovia) and fail 8% more often; spread from neighbours halved [9.3].	Enemy Repost & Rebrand
FRAMED	Enemy Frame the Story	2 weeks (3 if Donovia framed it)	Does not recover in upkeep; every enemy attack ×1.4.	Your Watch Team or Lessons Learned

8.3 HOW LONG A STATUS LASTS

A status counts the week it is set as its first week, and loses a week at each upkeep [9.5]. Setting it again on the same node starts the count again; it does not add.

8.4 COVERAGE

Watch Team and Prebunk cover the target **and every node linked to it**. The map draws each as one dashed outline with the weeks left. Pick the card and hover a node to see what it would cover.

8.5 LESSONS LEARNED

Each Lessons Learned raises your recovery by 1 for the rest of the campaign (Red's by 0.5), up to three times [9.1]. It also moves every node in your mind 4 toward you (Red's 2) and clears framing.

EXAMPLE

Olvana frames Blue's Cabinet in week 5. In week 5's upkeep it does not recover. In week 6 Blue plays a Watch Team next to it: the framing is cleared, and the Cabinet recovers in that week's upkeep. Lessons Learned would also have cleared it.

8.6 WARNINGS AND THE PROTECT VIEW

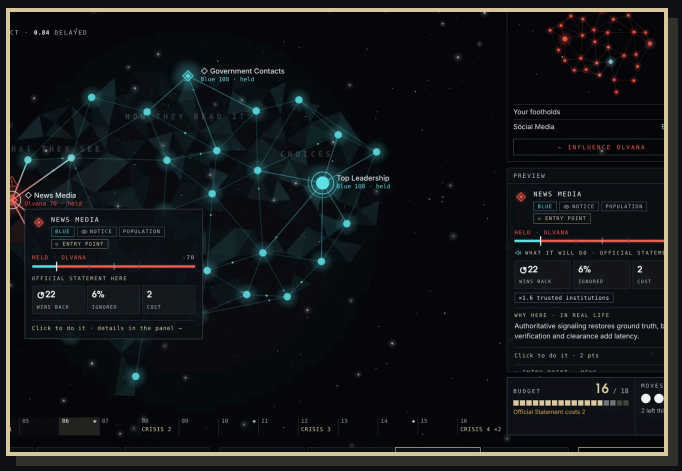


FIG. 11 PROTECT BLUE with Official Statement selected, hovering Blue's News Media, which Olvana now holds. The hover card shows it would win back 22 points.

The **PROTECT** tab shows your own mind; its ▲ count is how many of your nodes are under pressure. While you look at the enemy mind, the mini panel lists your three nodes in most trouble and suggests a card for each. When an area is one node from falling, the panel shows a red-ruled warning naming the penalty [5.5].

> PLAYER'S NOTE Two weeks before a crisis, defend the area it tests.

9.0 UPKEEP

After both turns, every node moves a little on its own. Upkeep works out every change from the board as it stands, then applies them all at once.

9.1 RECOVERY

Each node that is **not held by the enemy** and **not framed** moves toward its owner by the owner's recovery rate:

- 1, plus 1 for each Lessons Learned the owner has played (Red: plus 0.5 each) [8.5];
- halved while the owner's TRUST area is lost;
- Olvana's entry points ×1.2, Donovia's ×1.1 (censors at work);
- Blue's TRUST nodes recover 2 more (free press and courts);
- a Red node leaning Blue recovers at about a third of the rate (the truth sticks).

A node the enemy holds does not recover by itself: it must be pushed back.

9.2 SUPPORT

A node moves 1 more toward its owner if at least one linked node leans 50 or more toward the owner. This applies even to a held node, and never adds more than 1.

9.3 SPREAD

Each linked node that leans 50 or more toward the **other** side pushes the node 5 toward that side (6 in Fever in Port Sabre), or half that if the node is prebunked. There is no limit: three such neighbours push 15.

9.4 LIMITS

No node goes past −100 or +100.

9.5 STATUSES COUNT DOWN

After the moves above, every status loses one week [8.3].

EXAMPLE

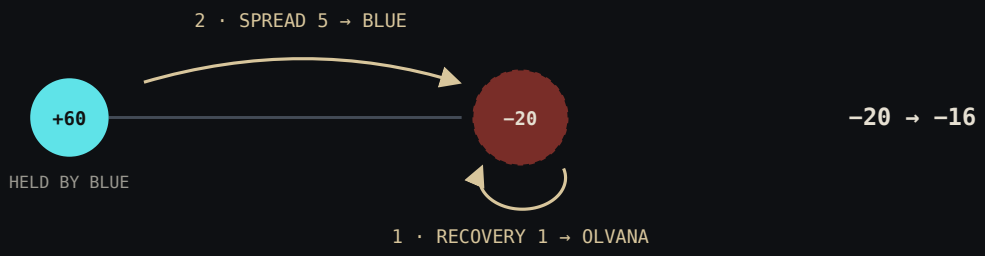
An Olvana node is at −20: it leans Olvana, but nobody holds it. At upkeep, two things happen to it:

1. It **recovers 1** toward Olvana, because it is not framed and Blue does not hold it. That makes −21.

2. A linked node that Blue holds strongly, at +60, **spreads 5** toward Blue. That makes **−16**.

Its other neighbours lean less than 50 either way, so they do nothing.

> PLAYER'S NOTE A foothold you ignore fades; a cluster you hold spreads.



The example above, drawn. Upkeep works out every change from the board as it stands, then applies them all at once.

SECTION 9 · REVIEW AND LOOKING AHEAD

9.6 THE WEEK IN REVIEW

The game explains itself at three moments: while Red moves, at the end of every week, and after the campaign [12.4].



FIG. C Week in review. Both sides' health, your moves and what each achieved, Red's moves with the most likely reason, how each of your areas moved, and what the next crisis needs.

9.6 WEEK IN REVIEW

After upkeep (and any crisis), the review opens. **CONTINUE** starts the next week; **SHOW ON MAP** lights up the nodes that changed first. It can be switched off [15.1].

9.7 WHILE RED MOVES



FIG. D Olvana's second move lights up the node it hit and the links it reached; the rest of the map dims. Its card says why it chose that target.

9.8 LOOKING AHEAD

From two weeks before a crisis, the side panel shows how each side would handle it today, how strong the tested area must be for SOUND, and where Red has been aiming. It judges Red only from moves you have seen.

PLAYER'S NOTE If Looking ahead says DELAYED, put this turn's defence into the tested area.

PLAYER'S NOTE The log keeps every move; ◉ opens its full story.

SECTION 10 · WHERE POINTS ARE WON

10.0 CRISES

A crisis is the moment a side must decide under pressure: a contaminated-water rumour, a severed cable, a fake evacuation alert.

10.1 WHEN

A crisis falls in each week marked CRISIS on the timeline: weeks 4, 8, 12 and 16 in the built-in campaigns. Each crisis tests one area, named in the top bar.

10.2 TIMING

Both sides are scored at the same moment: after upkeep [9.0], at the end of the crisis week.

10.3 THE SCORE

Each side scores 0.6 × its Decision Health + 0.4 × the health of the tested area. If that area is lost [5.4], subtract 0.08.

10.4 THE RESULT

SCORE	RESULT	POINTS
0.86 or above	SOUND	3
0.72 to under 0.86	DELAYED	1
Under 0.72	ERROR	0, and the other side gains 1

10.5 DOUBLE CRISES

A crisis marked ×2 doubles every point it awards, including the 1 from an enemy ERROR. In the built-in campaigns only the last crisis is double.

EXAMPLE

Week 8 tests INTERPRET. Blue's Decision Health is 0.84 and its INTERPRET area is at 0.80. Blue scores 0.6 × 0.84 + 0.4 × 0.80 = 0.824: DELAYED, 1 point. Olvana scores 0.69: ERROR, so Blue gains 1 more, 2 in all. In a ×2 crisis Blue would take 4.

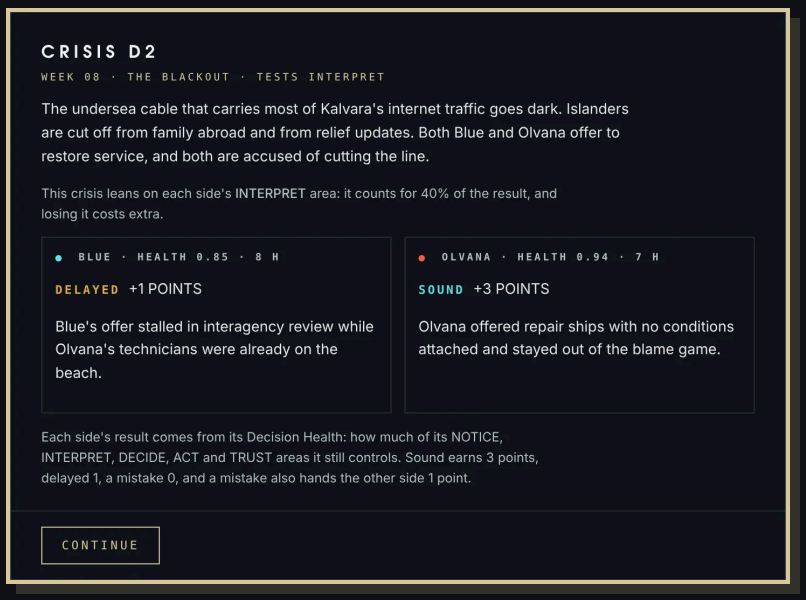


FIG. 15 Crisis 2, The Blackout. Blue was slow (DELAYED, 1 point); Olvana handled it soundly (3 points). The text says what each side actually did.

10.6 DECISION HEALTH

Each area has a **health** from 0 to 1: how much of it the owner still controls, with part credit for nodes that lean but are not lost. Decision Health adds up four of them by how much each side leans on them:

	NOT	INT	DEC	ACT
BLUE	20%	25%	30%	25%
OLVANA, DONOVIA	15%	20%	40%	25%

That total is multiplied by 0.75 + 0.25 × TRUST health, then 0.06 comes off for each lost area [5.5], and any flood penalty [13.4]. Difficulty never changes it. The side panel shows the sum.

> PLAYER'S NOTE The tested area counts twice, in Decision Health and again as 40% of the score.

SECTION 11 · WEEKS 2 · 6 · 10 · 14

11.0 NEW TECHNOLOGIES

Four times a campaign a new technology arrives. Each side can use it, or prepare for the other side using it.

11.1 THE CHOICE

In each new technology phase [4.1], both sides choose in secret: **USE IT** or **PREPARE DEFENSES**. Both choices are shown once both are in, and stay listed in the side panel.

11.2 EFFECTS

USE IT: for the rest of the campaign, the attacks the technology lists reach about ten times the audience (roughly one grade higher [7.6]), plus any special effect below.

PREPARE DEFENSES: the other side gets no bonus from this technology against you, and your three entry points are prebunked for 2 weeks.

EXAMPLE

Social Bots arrive in week 6. Blue uses them; Olvana prepares. Blue's Bot Swarm gets nothing from Social Bots, now or later. Olvana's three entry points are prebunked through week 7.

PLAYER'S NOTE Prepare when the other side leans on the cards a technology boosts.

11.3 THE SEVEN TECHNOLOGIES

TECHNOLOGY	BOOSTS
ALGORITHMIC TARGETING	Frame the Story +1 grade
SOCIAL BOTS	Bot Swarm +1 grade, full splash
TROLL FACTORIES	Bot Swarm, Repost & Rebrand +1 grade
AI SYNTHETIC MEDIA	Every attack +1 grade
HUMANOID ROBOTS	Plant Stories +1 grade, and it can reach any NOTICE or INTERPRET node, no foothold needed
AUTONOMOUS AI AGENTS	Bot Swarm, Repost & Rebrand +1 grade, full splash
INTERNET OF THINGS	Frame the Story, Repost & Rebrand +1 grade

Each built-in campaign uses four of these; the Olvana campaigns draw on the first five. The scoring method, the 1–3 scale and the threshold of 9 come from Rushing, Umeokolo & Xu (2025). The paper scores AI Synthetic Media and Humanoid Robots; the other score sets are the designer's estimates.

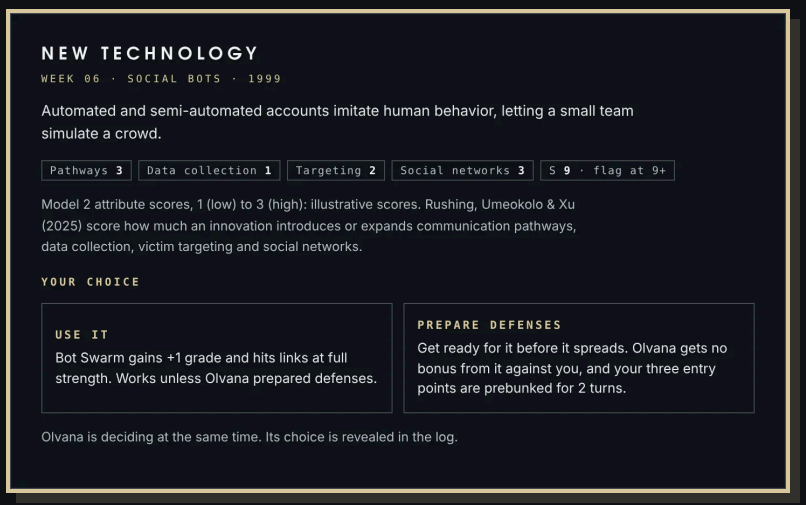


FIG. 14 Social Bots arrive in week 6. The four attribute scores sum to S = 9 (the designer's estimate), the paper's threshold for a flagged innovation.

12.0 WINNING

12.1 POINTS

After the last crisis, the side with more crisis points wins [10.4].

12.2 TIES

If points are level, the side whose Decision Health is more than 0.03 higher wins. Otherwise the game is a draw.

12.3 KNOCKOUT

At the end of each week, a side has the **upper hand** if the other side has at least 2 areas lost and at least 2 more lost areas than it does. A side that has the upper hand at the end of 2 weeks running wins at once. The side panel shows it as LEAD 0/2.

EXAMPLE

At the end of week 9, Olvana has lost DECIDE and INTERPRET; Blue has lost nothing. Blue has the upper hand: LEAD 1/2. If it still does at the end of week 10, Blue wins before week 11. If Olvana wins back INTERPRET in week 10, the count returns to 0.

12.4 THE DEBRIEF

When the game ends, the debrief gives a neutral look at both sides: health week by week, three ways to judge who led, the turning points and lessons for next time. The replay steps through the campaign week by week.

> PLAYER'S NOTE Play the same campaign twice, attacking first then defending first, and compare debriefs.

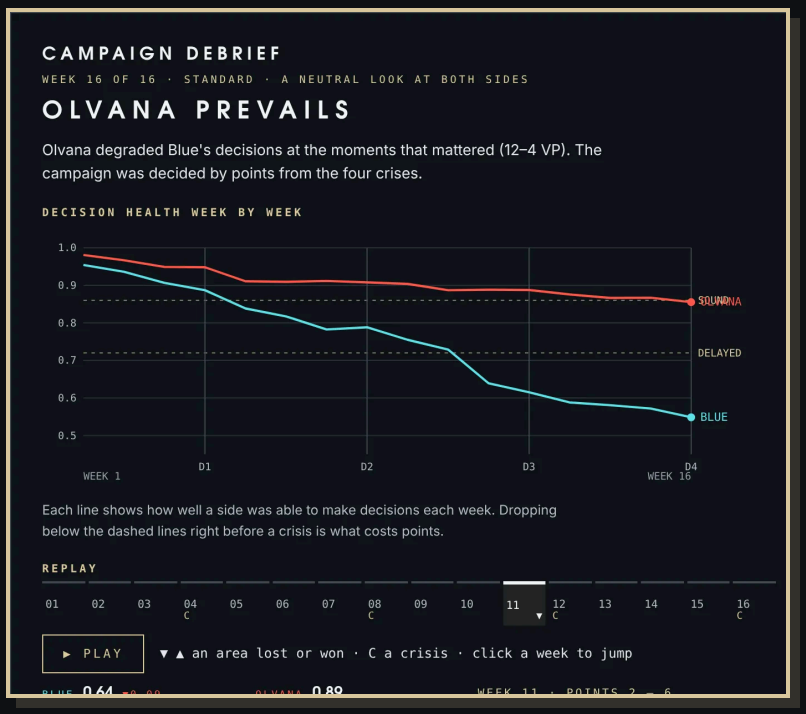


FIG. 16 The debrief: both sides' health over time, the verdicts and the turning points.

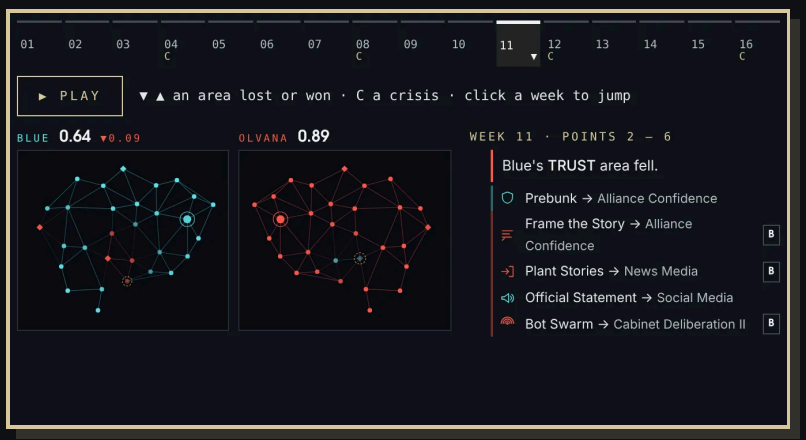


FIG. E The replay. Both minds week by week, the moves that week, and the turning points (▼ ▲) and crises (C) on the strip.

SECTION 13 · ASYMMETRIC SIDES

13.0 THE OPPONENTS

Blue is modelled as an open, rule-bound democracy; both opponents as closed, authoritarian states. The rules in sections 5 to 12 apply to every side except where this section says otherwise.

13.1 BLUE

OPEN INTERNET	Red's posts into Blue reach more people [7.7].
FREE PRESS & COURTS	Blue's TRUST nodes recover 2 more each week [9.1]; Official Statements push 22, not 14, while Blue's TRUST holds.
THE TRUTH STICKS	Red nodes leaning Blue recover at about a third of the normal rate [9.1].
CHEAP EXTRA MOVE	Blue's extra move costs 2 points [6.4].

13.2 BOTH OPPONENTS

ALREADY INSIDE	Blue's Social Media entry point starts at 80, not 100.
STATE-BACKED CONTENT	Bot Swarm costs 1 less.
RUN FROM THE TOP	Decision Health leans 40% on DECIDE (Blue's: 30%) [10.6].
SLOW TO LEARN	Lessons Learned heals 2 instead of 4 and adds 0.5 recovery, not 1 [8.5].
COSTLY EXTRA MOVE	An extra move costs 5 points.

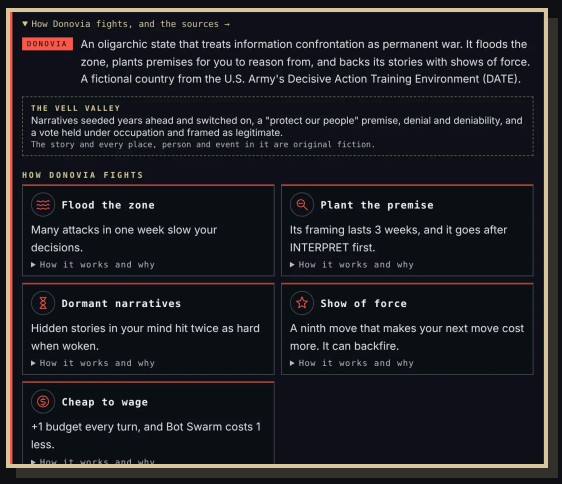


FIG. H How Donovia fights on the start screen: one card per rule, each with its source.

READING AN OPPONENT IN THE GAME

Pick a campaign and the start screen shows how its opponent fights, one card per rule, each with the source it comes from. The same cards are in the in-game guide.

> PLAYER'S NOTE Open on an entry point: every later attack needs the foothold it gives.

> PLAYER'S NOTE Both opponents lean on DECIDE; shaking it moves their health most.

> PLAYER'S NOTE Buy Lessons Learned early: faster recovery pays every week after.

13.3 HOW OLVANA FIGHTS

Olvana, a fictional country from the U.S. Army's training environment, is modelled as a closed, centralized one-party state that wages a long campaign through every arm of government. Each rule says where it comes from; design assumptions are marked as such.

HOW IT FIGHTS

CENSORED INTERNET

Your posts lose reach; censors delete what gets in.

Strongest firewall in the game: your posts into Olvana reach about 30% fewer people and fail 10% of the time instead of 8%, and its three entry points recover 20% faster.

DESIGN ASSUMPTION · ATP 3-13.1

STATE-FUNDED CONTENT

Content is cheap and plentiful.

Bot Swarm costs 1 less.

SHIN (NIU)

ALREADY INSIDE

Its personas are in place before the first crisis.

Your Social Media entry point starts at 80, not 100.

SHIN (NIU)

PATIENT AND METHODICAL

Steady pressure, not bursts.

The computer works one of your areas at a time, varies its cards and rarely buys an extra move.

SHIN (NIU) · DESIGNER'S READING

WHERE IT IS WEAK

RUN FROM THE TOP

DECIDE is its single point of failure.

Health leans 40% on DECIDE (Blue's: 30%). Fast when intact, brittle when shaken.

SHIN (NIU)

HARD TO ADMIT MISTAKES

Bad news travels up slowly.

Lessons Learned heals +2 instead of +4 and builds recovery half as fast.

DESIGN ASSUMPTION

COSTLY EXTRA MOVES

It cannot improvise cheaply.

An extra move costs Olvana 5 points; Blue pays 2.

DESIGN ASSUMPTION

Shared with Donovia [13.2]: already inside your Social Media entry point, cheaper Bot Swarm, run from the top, slow to learn, extra moves cost 5.

PLAYER'S NOTE Hit an Olvana entry point early, before its censors heal it.

PLAYER'S NOTE Guard your own Social Media: it starts weak against Olvana.

PLAYER'S NOTE Once DECIDE cracks, Olvana learns too slowly and pays too much to surge its way out.

SECTION 13 · KNOW YOUR OPPONENT

13.4 HOW DONOVIA FIGHTS

Donovia, a fictional country from the U.S. Army's training environment, is modelled as an oligarchic state at permanent information war. Its rules are drawn from published research; the numbers are the designer's.

HOW IT FIGHTS

FLOOD THE ZONE

Many attacks in one week slow your decisions.

More than 2 attacks landing in a week: each extra one takes 0.02 off your health until Donovia's next turn (up to 0.06). Each Lessons Learned level absorbs one more.

NATO C2COE · ISW · WILDE & SHERMAN

PLANT THE PREMISE

Its framing lasts longer, and it aims at how you read events.

Frame the Story keeps a node framed 3 weeks, not 2. The computer goes for INTERPRET first, then DECIDE.

NATO C2COE · ISW · IISS

DORMANT NARRATIVES

Stories seeded long ago, hidden in your mind.

3 hidden in your NOTICE, INTERPRET and TRUST nodes (more in some campaigns). The first attack on one hits twice as hard; a Watch Team removes it.

ISW

SHOW OF FORCE

A ninth move: an act, not a post.

3 points: any ACT or DECIDE node, no foothold, no audience roll. Your next move costs 1 more. Failure backfires on the node and its neighbours.

ISW · IISS

CHEAP TO WAGE

Its information war costs little.

+1 budget every turn; Bot Swarm costs 1 less. The computer repeats cards more readily and buys extra moves often.

ISW

WHERE IT IS WEAK

PREDICTABLE, BRITTLE

Its messages are predictable and its threats can backfire.

Prebunk works better against it (×0.45), its failed attacks push back harder, and your attacks on its TRUST area hit 10% harder.

ISW

LEAKY FIREWALL

Its internet is harder to seal than Olvana's.

Your posts lose about a fifth of their reach (about 30% into Olvana) and fail 9% of the time; its entry points recover only 10% faster.

WILDE & SHERMAN

ONE COMMAND

DECIDE is its single point of failure.

Health leans 40% on DECIDE, as Olvana's does.

NATO C2COE

Shared with Olvana [13.2]: already inside your Social Media entry point, slow to learn, extra moves cost 5. Show of Force is key 9.

PLAYER'S NOTE Watch Teams on NOTICE, INTERPRET and TRUST flush out dormant narratives.

PLAYER'S NOTE Defend the premise the next crisis tests, not every single story.

14.0 TWO PLAYERS

Two people can play one campaign, one as Blue and one as Red, by swapping short lines of text. Send them any way you like: email, chat, Teams.



FIG. 17 TWO PLAYERS on the start screen. Blue starts a game or a structured session; Red joins by pasting Blue's first code, and a White Cell by pasting Red's. Saved games appear under RESUME.

14.1 TURN CODES

Each turn ends with a **turn code**: a line of text that carries your moves, your new-technology choice, the game's ID, starting seed, campaign and difficulty, and a fingerprint of the board. It carries nothing personal and nothing from your computer. The other player pastes it in and watches your moves replay.

> PLAYER'S NOTE Neither side sees the other's moves until the code arrives.

14.2 BLUE STARTS

1. Pick a campaign and difficulty. The campaign decides whether Red is Olvana or Donovia.
2. **TWO PLAYERS → NEW GAME AS BLUE**.
3. Make your moves and press **END TURN · GET CODE**.
4. **COPY CODE** and send it. The first code is also the invitation.

14.3 RED JOINS

1. **TWO PLAYERS**, paste Blue's code, **JOIN AS RED**.
2. Blue's moves replay; then it is your turn.
3. End your turn, copy your code and send it back.

14.4 EVERY WEEK AFTER

Paste the other player's code, play, send yours. Repeat until the game ends.

14.5 THE LAST CODE

When Red's last turn ends the game, Red gets one final code to send so Blue's game ends too. In a structured session that code goes to the White Cell, who sends Blue the last one [14.6].

14.6 STRUCTURED SESSIONS

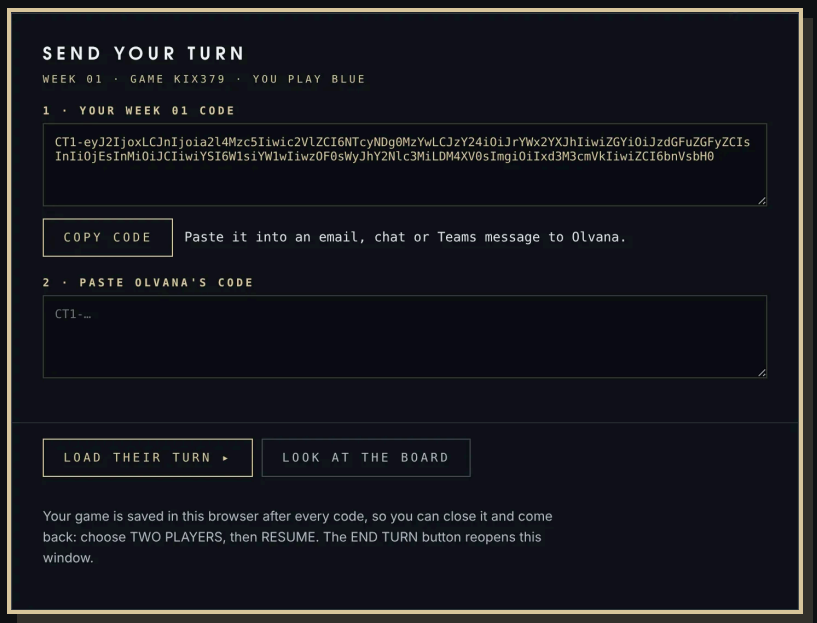


FIG. 18 After END TURN: your code at the top with a COPY button, and a box for the reply below. LOOK AT THE BOARD closes the window; the END TURN button (now TURN CODES ▸) brings it back.

14.6 THE WHITE CELL

A facilitator can run the game as the **White Cell**. Blue starts with **NEW STRUCTURED SESSION · WITH WHITE CELL**; the facilitator joins with Red's first code (**JOIN AS WHITE CELL**). Codes then go Blue → Red → White Cell → Blue.

Each week the White Cell may run stories, make rulings and change the board. A **WHITE CELL** notice lists them: Blue sees it before its turn, Red when Blue's code arrives. Follow its rulings. A move it strikes replays without it.

14.7 SAVING AND RESUMING

A two-player game saves after every code. To go back to it, open **TWO PLAYERS** and pick it under **RESUME**. The tutorial is single-player only; your two-player game stays saved while you take it.

14.8 VERSIONS

You may mix the Windows app, the Mac app and the HTML file if every player is on the same minor version (0.17, 0.17.1 ...). The game cannot check this for you. For a player-made campaign, every player imports it first.

14.9 BOARD CHECK

Each code carries a fingerprint of the board. If the log says **Board check failed**, the copies disagree: compare health bars, and restart from the last code you both agree on.

14.10 WHAT CHANGES FROM SINGLE PLAYER

- Whoever plays Red sees the game from Red's side: their mind is "yours", and the tabs read PROTECT OLVANA and INFLUENCE BLUE (or DONOVIA).
- Your own moves get the large result cards; your opponent's appear as small cards at the top of the screen.
- New-technology choices stay secret until both are in.
- There is no week in review; use the log and the debrief.
- Immersive Briefings are single-player only [15.3].

The game needs the internet only to send codes; it runs offline.

> PLAYER'S NOTE For a seminar: pair people up, give them a week, debrief together.

SECTION 15 · ⚙ IN THE TOP BAR

15.0 SETTINGS

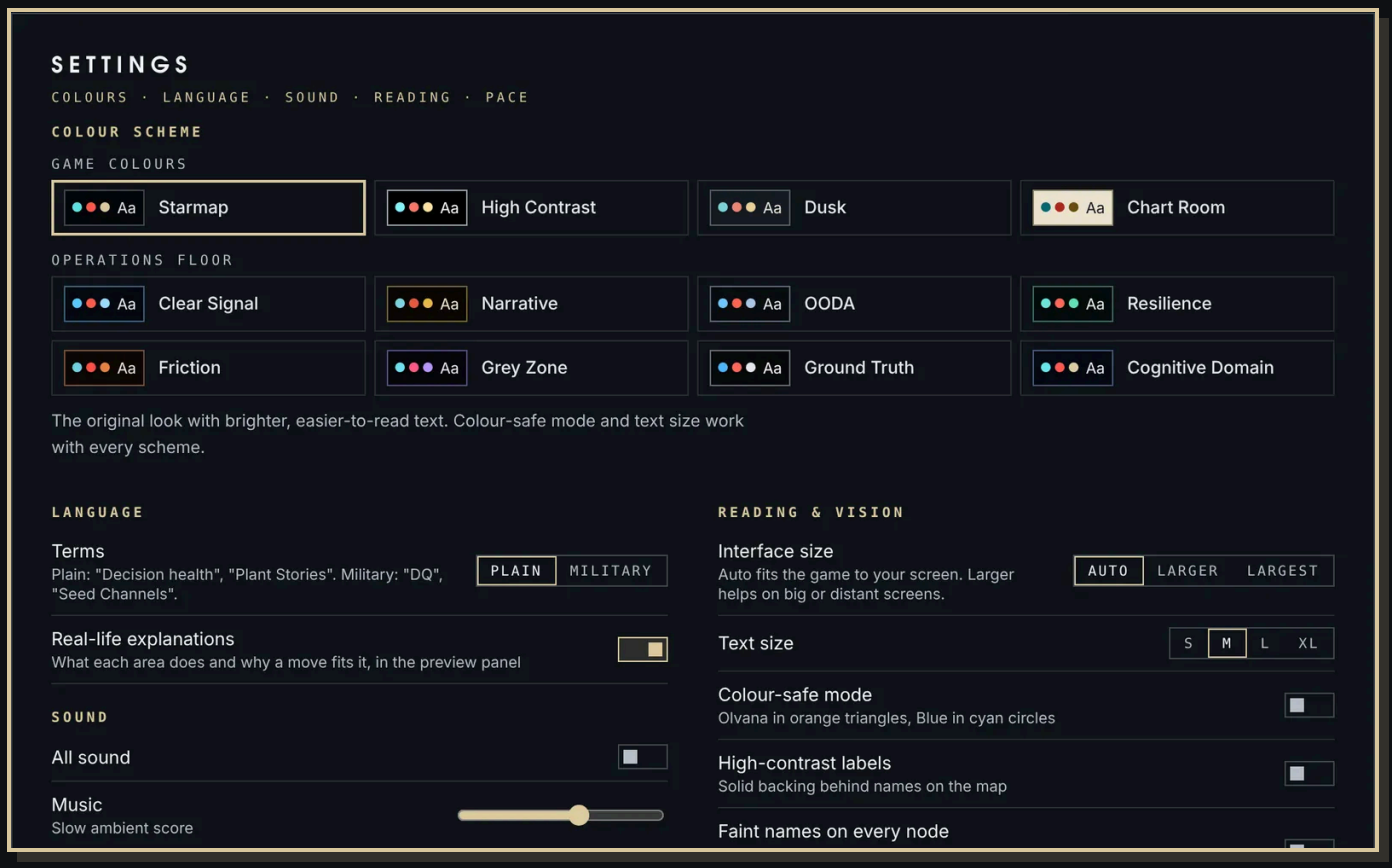


FIG. 19 Settings: colour schemes, language, real-life explanations, sound and music, interface and text size, colour-safe mode, map labels, the decision-loop overlay, reduced motion, Red's turn speed and how often result cards pop up. Immersive Briefings: 15.3.



FIG. 20 Colour-safe mode: Red becomes orange triangles, Blue stays cyan circles, so the sides differ by shape as well as colour.

15.1 ACCESSIBILITY AND DISPLAY

Settings change how the game looks and reads, never its rules. You can choose one of twelve colour schemes, enlarge the interface and the text, switch on colour-safe mode, show every node's name, overlay the decision loop, reduce motion, slow Red's turn down, and choose how often result cards and the week in review appear. Plain or military language is here too [3.4].

15.2 SOUND AND MUSIC

Choose one of four music tracks: Night Watch, Undertow, Long Telegram or Dead Air. The music and sound buttons sit in the top bar.

> PLAYER'S NOTE Every setting is kept in the browser or app you play in.

SECTION 15 · OPTIONAL NARRATION

15.3 IMMERSIVE BRIEFINGS

A free, optional voice pack. With it, each week in review opens with a short broadcast, read aloud by the side that is ahead. Without it, nothing in the game changes.

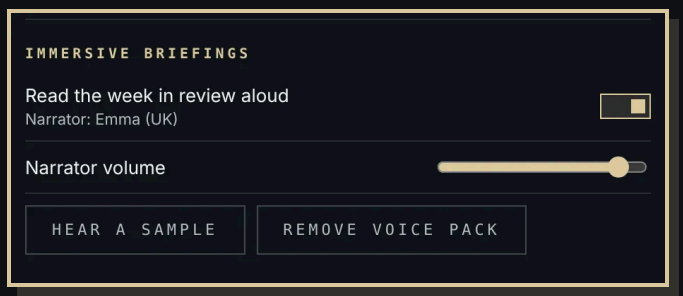


FIG. 21 The broadcast at the top of the week in review. Blue is ahead, so Blue Public Broadcast tells the week; the line being read is lit, like a caption. When Red leads, its own state channel tells the same week its way.

15.4 GETTING THE PACK

1. Download **CognitiveTerrain-Voice-Emma.zip** (about 111 MB) from the playtest page, or from Settings › Immersive Briefings › **GET THE PACK**.
2. **Browser:** Settings › Immersive Briefings › **LOAD VOICE PACK...** and choose the zip, or drop it on that box. The browser keeps it.
3. **Windows, macOS, Linux:** put the zip beside the game before you start it, or load it from Settings the same way.

Leave it zipped. The switch turns the voice off; **REMOVE VOICE PACK** deletes it.



Settings › Immersive Briefings with the pack loaded.

15.5 QUESTIONS

Is the voice real? No. It is synthetic speech ("Emma, UK") made on your own computer by an open-source text-to-speech model, Kokoro-82M. It works offline; nothing is sent anywhere.

Is it safe? The game checks every file in the pack against fingerprints built into the game and refuses anything changed.

What does it say? It is built from the week's real moves: the leader's best story, the other side's failures, areas won or lost, the next crisis.

Slow computer? The text shows at once; the voice waits until it can read to the end without gaps. Continue whenever you like: it plays on, with an ON AIR tab to stop it.

Two players? Single-player only.

REFERENCE TABLES

Every number in one place. Each table points back to the rule it comes from.

CARDS [7.2] [8.1] [13.4]

CARD	COST	PUSH	COVERS
PLANT STORIES	2	20 (×1.5 on entry)	Target
FRAME THE STORY	2	16; framed	Target
BOT SWARM	3*	12; 7 on links	Target + links
REPOST & REBRAND	2	10; strips watched, prebunked	Target; strips on links too
SHOW OF FORCE	3	16 (no grade)	Any ACT/DECIDE
WATCH TEAM	2	Watched 3 wk	Target + links
PREBUNK	2	Prebunked 3 wk	Target + links
OFFICIAL STATEMENT	2	14 back (Blue 22)	Target
LESSONS LEARNED	3	+4 to all (Red +2)	Whole mind

* 2 for Red. Add 1 to every cost while INTERPRET is lost.

FAILURE [7.5]

Attack	8%, +15% watched, +8% prebunked
Blue into Olvana / Donovia	+2% / +1%
Show of Force	+4%
Official Statement	6%; 26% if TRUST lost
Any move, ACT lost	1 in 4 misfires

GRADES [7.6]

SCORE	GRADE	PUSH ×
under 3	F	0.25
3 to under 4	E	0.45
4 to under 10	D	0.7
10+	C	1
100+	B	1.2
1,000+	A	1.4
10,000+	A+	1.6

CRISES [10.3] [10.4]

Score = 0.6 × Decision Health + 0.4 × tested area, −0.08 if that area is lost

SOUND	0.86 or above: 3 points
DELAYED	0.72 to under 0.86: 1 point
ERROR	under 0.72: 0, other side +1

WEEKLY RECOVERY [9.1]–[9.3]

Recovery	1 + Lessons level (Red +0.5 each)
Support	+1 if a link leans 50+ your way
Spread	5 per link leaning 50+ the other way

THRESHOLDS [5.2]–[5.4] [6.5]

Foothold	10	Held	30
Area lost	half its nodes	Budget cap	18

GLOSSARY

ON SCREEN: PLAIN · MILITARY

PLAIN	MILITARY
Decision Health	Decision Quality (DQ)
Budget · pts	Influence points · IP
Notice / Interpret	Observe / Orient
Influence / Protect	Shape / Defend
Top Leadership	National Command
Entry point	Gateway
Lost (area)	Compromised
Watched	Detected
Prebunked / Framed	Inoculated / Primed
Crisis	Decision point
Lead	Superiority
New technologies	Disruptive innovations
How far it spread	Engagement effectiveness
What happened	SITREP
Plant Stories	Seed Channels
Frame the Story	Frame & Prime
Bot Swarm	Coordinated Amplification
Repost & Rebrand	Pivot & Reseed
Watch Team	Assessment Cell
Prebunk	Prebunk & Inoculate
Official Statement	Authoritative Response
Lessons Learned	After-Action Review

In military mode Red's moves also use its own doctrine names, such as "Public Opinion Warfare".

TERMS IN THIS MANUAL

TERM	PLAIN MEANING · IN THE GAME
INFORMATION OPERATIONS	Using information to affect what an opponent knows, believes and decides, while protecting your own side. → The whole game
COGNITIVE WARFARE	A long contest over how people perceive, interpret, decide and act. → The whole game
OODA LOOP	Observe, Orient, Decide, Act: the four steps of any decision. → NOTICE, INTERPRET, DECIDE, ACT
MISO / PSYOP	Military Information Support Operations: truthful messages aimed at foreign audiences. → Plant Stories, Frame the Story
PUBLIC AFFAIRS	Official communication with the public and press. → Official Statement
OPSEC	Protecting information an opponent could exploit. → Watch Team
PREBUNKING	Teaching people a manipulation trick before they meet it. → Prebunk
AMPLIFICATION	Making a message louder: bots, troll networks, coordinated sharing. → Bot Swarm
SYNTHETIC MEDIA	AI-generated text, images, audio or video. → New technology events
FRAMING	The angle a story is told from. → Frame the Story
ATTRIBUTION	Deciding, and saying publicly, who was responsible. → INTERPRET area
ENGAGEMENT EFFECTIVENESS	A research score for user interactions per post: shares count most, then comments, likes, views. → Spread grade F to A+
COGNITIVE SUPERIORITY	Staying ahead over time: deciding well, adapting faster, getting results cheaper. → LEAD; the knockout win
ACUTE vs CHRONIC	Fast effects (minutes to days) versus slow ones (weeks to years). → Most areas vs TRUST
REFLEXIVE CONTROL	Shaping what an opponent sees so it reaches, by itself, the decision you want. → Donovia's Frame the Story
ODIN / DATE / OPFOR	US Army training environment and the fictional countries and opposing forces it defines. → Olvana, Donovia and their neighbours

APPENDIX C · PLAYER AID · SIDE 1 OF 2

SEQUENCE OF PLAY

EACH WEEK [4.0]

- **New technology** (weeks 2, 6, 10, 14): USE IT or PREPARE DEFENSES, in secret [11.0].
- **Blue's turn**, then **Red's turn** (see Your turn).
- **Upkeep**: nodes recover, strong nodes spread, statuses lose a week [9.0].
- **Crisis** (weeks 4, 8, 12, 16): both sides scored [10.0].
- **Week in review**, then the next week begins.

THE SIXTEEN WEEKS · ◆ TECHNOLOGY · C CRISIS															
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	◆		C		◆		C		◆		C		◆		C×2

YOUR TURN [6.0]

1. **Collect income**: 6, +1 for each enemy node you hold (Top Leadership 2), +2 more for their Government Contacts; never under 2. You start with 8. Anything over 18 is lost.
2. **Pick a card** (keys 1–8; 9 is Show of Force). Most cost 2; Bot Swarm and Lessons Learned 3; +1 while INTERPRET is lost.
3. **Click a ringed node** to make the move. Up to 3 moves a turn (2 while DECIDE is lost). Esc cancels.
4. **Extra move**, if you want it: once, after your last move. Blue pays 2, Red 5, and must keep 2.
5. **END TURN.**

CONTROL [5.0]

LEAN	−100 (Red) to +100 (Blue)
FOOTHOLD	10+ your way: reach its links
HELD	30+ your way: income; it no longer recovers
AREA LOST	Half its nodes held (round up); DECIDE also if Top Leadership is held

REACH [7.1]

Entry points ◇, footholds, and nodes linked to a foothold. Defences: your own mind.

HOW HARD IT LANDS [7.4]

Start with the card's push. It grows in the card's best areas and with a bigger audience (grade), and Plant Stories hits 50% harder on an entry point. A prebunked node takes half; a framed node takes 40% more. *Example: Plant Stories on News Media, grade B: 20 → 30 → 45 → 54.*

CRISIS [10.0]

0.6 × Decision Health + 0.4 × tested area

−0.08 IF THE TESTED AREA IS LOST

SOUND	0.86+	3 points
DELAYED	0.72 to under 0.86	1 point
ERROR	under 0.72	0; other side +1

×2 crises double every point. Most points wins; level points: Decision Health more than 0.03 higher wins, else a draw [12.2].

KNOCKOUT [12.3]

The other side has 2+ areas lost and 2+ more than you, at the end of 2 weeks running.

APPENDIX C · PLAYER AID · SIDE 2 OF 2

CARDS AND AREAS

CARD	KEY	COST	DOES (BASE PUSH)	BEST IN
PLANT STORIES	1	2	Push 20; ×1.5 on entry points	NOTICE
FRAME THE STORY	2	2	Push 16; no recovery for 2 turns	INTERPRET, DECIDE, TRUST
BOT SWARM	3	3	Push 12 + 7 on every link	ACT, TRUST
REPOST & REBRAND	4	2	Push 10; strips Watched and Prebunked	Anywhere
WATCH TEAM	5	2	3 weeks: reach ÷10, statements ×1.5	Anywhere
PREBUNK	6	2	3 weeks: half influence on node + links	Anywhere
OFFICIAL STATEMENT	7	2	Restore 14 (Blue 22); 7 once TRUST is lost	Anywhere
LESSONS LEARNED	8	3	All your nodes +4; faster recovery (max 3)	Anywhere
SHOW OF FORCE	9	3	Donovia only. Push 16 (24 on ACT, 20 on DECIDE), no foothold; Blue's next move +1	ACT, DECIDE

LOSE THIS AREA…	…AND THIS HAPPENS
NOTICE	You can no longer see the other brain clearly
INTERPRET	Every move costs +1
DECIDE	−1 move per turn (also if Top Leadership falls)
ACT	1 in 4 moves fails
TRUST	Statements fall to 7, recovery halves, enemy reach ×10

ENTRY POINT Always in reach

LARGE NODE Top Leadership

WATCHED Attacks land weaker

FRAMED Will not recover yet

THIS WEEK'S ORDERS Above the timeline; click one

CONTESTED Nobody holds it

▲ ON YOUR BRAIN Under pressure

PREBUNKED Pushes count half

DASHED OUTLINE Nodes one defence covers

ON AIR Immersive Briefings is speaking

STATUS	LASTS	WHAT IT MEANS
WATCHED	3 wk	Enemy posts reach few people and often fail. Your statements hit harder.
PREBUNKED	3 wk	Enemy attacks push half as hard and fail more often.
FRAMED	2 wk (Donovia 3)	Can't recover. Enemy attacks hit harder.

FAIL CHANCE	
Attack	8%
Blue into Olvana / Donovia	10% / 9%
Official Statement	6% (26% TRUST lost)

SOURCES AND CREDITS

Built on published research; the rules simplify it.

REFERENCES

- Rushing, B., Hersch, W., & Xu, S. (2026). *Cognitive Warfare: Definition, Framework, and Case Study.* arXiv:2603.05222.
- Rushing, B., & Xu, S. (2025). *Quantifying the Engagement Effectiveness of Cyber Cognitive Attacks: A Behavioral Metric for Disinformation Campaigns.* arXiv:2510.15805.
- Rushing, B., Umeokolo, M.-R., & Xu, S. (2025). *Towards Proactive Defense Against Cyber Cognitive Attacks.* arXiv:2510.15801.
- Headquarters, Department of the Army (2018). *ATP 3-13.1, The Conduct of Information Operations.*
- NATO Science & Technology Organization. *Cognitive Warfare: NATO Chief Scientist Research Report.*
- Shin, D. W. (2026). *Viewing China's Cognitive Warfare Through the Lens of Its Global Strategy.* National Intelligence University.
- Eads, L. J., Clarke, R., Lin, X. S., & McCreight, R. (2023). *Warfare in the Cognitive Age: NeuroStrike and the PLA's Advanced Psychological Weapons & Tactics.*
- NATO Command and Control Centre of Excellence (2026). *Information Operations in Russian Command and Control* (short read).
- Bugayova, N., & Stepanenko, K. (2025). *A Primer on Russian Cognitive Warfare.* Institute for the Study of War.
- Voo, J., & Singh, V. V. (2025). *Russia's Information Confrontation Doctrine in Practice (2014–Present): Intent, Evolution and Implications.* International Institute for Strategic Studies.
- Wilde, G., & Sherman, J. (2023). *No Water's Edge: Russia's Information War and Regime Security*. Carnegie Endowment for International Peace.
- U.S. Army TRADOC G-2. *Decisive Action Training Environment (DATE)*, Operational Environment Data Integration Network (ODIN): Olvana, Donovia, Bothnia, Framland, Arnland, Gorgas. odin.t2com.army.mil/DATE
- Roozenbeek, J., & van der Linden, S. (2020). *Breaking Harmony Square: A game that inoculates against political misinformation* (cited in Rushing et al.).

ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

Changes made: the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; 16 of the 18 technology score sets are the designer's own estimates (the paper scores AI Synthetic Media and Humanoid Robots); all scenario, narrative and tutorial text is original.

No endorsement: the authors and their institutions have not reviewed or endorsed this game.

Immersive Briefings voice pack (optional, separate download): Kokoro-82M by hexgrad, kokoro-js and transformers.js (Apache 2.0); ONNX Runtime (MIT); eSpeak NG (GPL-3.0-or-later), whose source is published with the pack.

HOW THIS GAME WAS MADE

Design and direction: Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

Tools: I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced.

The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image or music generators were used, and the base game runs no AI while you play. The optional Immersive Briefings voice is synthetic speech from an open-source model that runs on your own computer (15.3).

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.